Spam Email Detection using Logistic Regression

1■■ Data Import and Preparation

```
from google.colab import files
uploaded = files.upload()

import zipfile, io
zip_name = list(uploaded.keys())[0]
with zipfile.ZipFile(io.BytesIO(uploaded[zip_name]), 'r') as zip_ref:
    zip_ref.extractall()

import pandas as pd, os, random
for file in os.listdir():
    if file.endswith('.csv') or file.endswith('.data'):
        dataset_file = file
        break

data = pd.read_csv(dataset_file, header=None)
```

Output:

```
spambase.zip uploaded successfully
```

2■■ Data Splitting and Scaling

```
X = data.iloc[:, :-1]
y = data.iloc[:, -1]

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_st
```

Output:

```
Training and test sets created successfully (80% train, 20% test)
```

3■■ Model Training and Evaluation

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model = LogisticRegression(C=2.0, solver='saga', max_iter=3000, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", round(accuracy * 100, 2), "%\n")
```

Output:

```
Model Accuracy: 91.97 %
```

4■■ Predicted vs Actual Comparison

```
print("Predicted vs Actual Results (1 = SPAM, 0 = NOT SPAM):\n")
compare_df = pd.DataFrame({'Actual': y_test.values, 'Predicted': y_pred}).reset_index(dro
sample_rows = compare_df.sample(10, random_state=42)
print(sample_rows.to_string(index=False))
```

Output:

```
 Actual  Predicted
      0          0
      1          1
      0          0
      1          1
      0          0
      1          1
      0          0
      0          0
      1          1
      1          1
```

5■■ Visualization Section

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

plt.figure(figsize=(14, 10))
plt.subplot(2, 2, 1)
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False)
plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')

plt.subplot(2, 2, 2)
plt.bar(['Accuracy'], [accuracy * 100], color=['green'])
plt.ylim(0, 100)
plt.title('Model Accuracy (%)')
plt.ylabel('Accuracy %')
plt.text(0, accuracy * 100 + 1, f"{round(accuracy * 100, 2)}%", ha='center', fontsize=12)

plt.subplot(2, 1, 2)
plt.plot(y_test.values[:50], label='Actual', color='blue', marker='o')
plt.plot(y_pred[:50], label='Predicted', color='red', linestyle='--', marker='x')
plt.title('Actual vs Predicted (First 50 Emails)')
plt.xlabel('Sample Index')
plt.ylabel('Spam(1) / Not Spam(0)')
plt.legend()
plt.tight_layout()
plt.show()
```

Output (Graphs):

- Confusion Matrix
- Model Accuracy Bar Graph
- Actual vs Predicted Plot

6■■ Random Email Predictions

```
print("\nPrediction on 5 Random Emails from Full Dataset:\n")
for i in random.sample(range(len(data)), 5):
    sample = scaler.transform([X.iloc[i]])
    pred = model.predict(sample)[0]
    print(f"Email {i}: {'SPAM' if pred == 1 else 'NOT SPAM'}")
```

Output:

```
Email 2497: NOT SPAM
Email 3462: NOT SPAM
Email 3128: NOT SPAM
Email 2571: NOT SPAM
Email 934: SPAM
```

■ Conclusion

The Logistic Regression model successfully detects spam emails with an accuracy of 91.97%. Visualization plots confirm correct model performance with high accuracy and minimal misclassifications.